

Refractive index and absorption spectra changes induced by UV irradiation in lead silicate glasses

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Abstract

The negative refractive index changes were obtained in lead silicate glasses with different lead concentrations (from 30 mol% PbO to 50 mol% PbO) by irradiation with the frequency-quadrupled output of a Q-switched YAG laser (266 nm, 10 Hz repetition rate) at low energy density (50 mJ/cm²). The largest refractive index change Δn was -0.25 ± 0.04 in lead silicate glasses with 50 mol% PbO. UV-visible absorption spectra of the lead silicate glass with 43 mol% PbO were measured after exposure to the 266 nm laser beams with energy density from 50 mJ/cm² to 350 mJ/cm². The absorption coefficients in visible wavelength increased suddenly when the energy density of the laser beam was larger than a threshold value. This showed that the structural changes of lead silicate glasses were different after exposure to the 266 nm laser beam with different energy densities.

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1. Introduction

In 1978, Hill et al. [1] first found the permanent refractive index change in germanosilicate glass fiber after optical irradiation. This phenomenon, which is called glass photosensitivity, allows the fabrication of many phase structure photonic devices which have many applications in telecommunications and other opto-electronics industries [2,3]. The preponderance of the investigations to date has been devoted to GeO₂-doped silica glass [4,5]. The maximum photo-induced index change Δn , the index difference before and after UV irradiation, is about 0.01 for hydrogen-loaded germanosilicate glass fibers with limited stability at elevated tem-

perature [6]. Recently, many other silicate glasses, such as tin-doped and lead-doped silicate glasses, were found to have very strong photosensitivity [7–9]. Long and Brueck wrote permanent gratings in lead silicate glasses through a phase mask by irradiation with a pulsed KrF excimer laser (248 nm) and a Q-switched YAG laser (266 nm), respectively [7,8]. The index change Δn can be evaluated by measuring the diffraction efficiency of the gratings. The largest index change ($\Delta n = 0.21 \pm 0.04$ at 633 nm) was obtained in SF59 glass (57 mol% PbO). But this method cannot distinguish that the Δn is positive or negative.

We have studied the UV-visible absorption spectra changes of lead silicate glasses with different composition after exposure to the 266 nm laser beam and found that the absorption spectra changes were different after exposure to the 266 nm laser beam with low (50 mJ/cm²) and high (150 mJ/cm²) energy density [10]. In this

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paper, we measured the refractive index of lead silicate glasses with different composition and found that the refractive index of lead silicate glasses decreased after exposure to the 266 nm laser beam with the energy density 50 mJ/cm^2 . The largest refractive index change Δn was -0.25 ± 0.04 in lead silicate glasses with 50 mol% PbO. We also studied the absorption spectra changes of lead silicate glass after exposure to the 266 nm laser beam with the energy density from 50 mJ/cm^2 to 350 mJ/cm^2 .

2. Experimental

The test samples were fabricated by a standard fusion method. The compositions of the samples are listed in Table 1. The samples were polished to plane parallel plates of 1.4 mm thickness. The optical quality of the plate surface allowed measurement of the absorption coefficient over the wavelength range 190–700 nm.

The irradiation source was the frequency-quadrupled output of a Q-switched YAG laser at 266 nm. The width of the pulse was about 10 ns and the repetition rate was 10 Hz. The diameter of the beam spot was about 5 mm. The pulse energy we used to irradiate the samples was 10–70 mJ, yielding an incident energy density of about 50–350 mJ/cm^2 .

We measured the refractive index of bulk lead silicate glasses by Brewster's angle. The experiment setup is shown in Fig. 1. The incident light was a p-polarized He-Ne laser beam. The reflective light was accepted

by a photodiode. The samples were laid on a precise rotator controlled by a computer. The computer collected a group of light intensity and angle data at each 0.004° . The smallest light intensity was corresponding to the Brewster's angle θ_B . According to the Brewster rule, the index n of the lead silicate glass is equal to $t_g \theta_B$. In order to improve our index measurement accuracy, we only collected data of light intensity and angle at the scope of 7° nearby Brewster's angle. The measurement accuracy of refractive index is about ± 0.02 .

We also studied the UV-visible absorption spectra changes of lead silicate glass with 43 mol% PbO after exposure to 266 nm laser with the energy densities from 50 mJ/cm^2 to 350 mJ/cm^2 . The irradiation time was 1 min. All absorption measurements were carried out at room temperature (20°C) using an UV-3101PC spectrophotometer (Shimadzu Corp.). The absorption coefficients were corrected for Fresnel reflection.

3. Results

Fig. 2 is the relationship of index and irradiation time of lead silicate glass with 50 mol% PbO after exposure to the 266 nm laser with the energy density 50 mJ/cm^2 . It can be seen from Fig. 2 that the index of lead silicate glass decreases after exposure to the 266 nm laser with the energy density about 50 mJ/cm^2 . The index decreases fast at the beginning of irradiation, then the index decreases slowly and become saturated after 30 min. Table 1 lists the index changes of four lead silicate glasses with different composition after exposure to the 266 nm laser with the energy density 50 mJ/cm^2 . The index changes of lead silicate glasses increase with the increasing of PbO content. Fig. 3 shows the relationship between refractive index changes Δn and the PbO contents. This result is corresponding to [7].

The surface of lead silicate glasses reveals brown spot after exposure to the 266 nm laser with high energy

Table 1

The refractive index changes (Δn) of lead silicate glasses with different composition after exposure to the 266 nm laser with low energy density (50 mJ/cm^2) (the irradiation time is 30 min)

Content	Sample number			
	1	2	3	4
PbO (mol%)	30	33	43	50
SiO ₂ (mol%)	70	67	57	50
Refractive index changes Δn	-0.09	-0.13	-0.17	-0.25

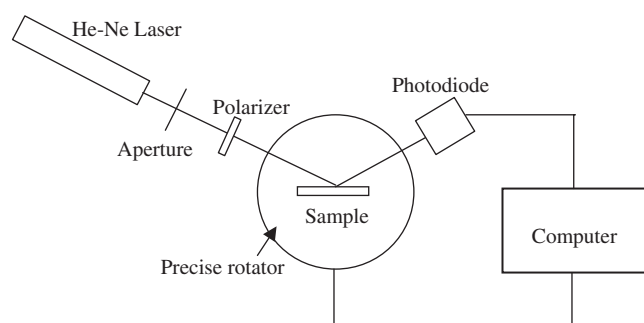


Fig. 1. The experiment setup of refractive index measurement by Brewster's angle.

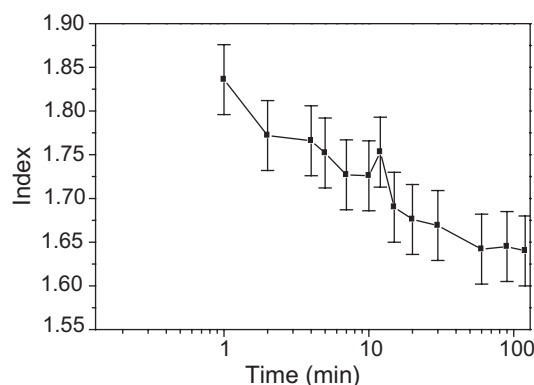


Fig. 2. The relationship between refractive index and irradiating time of lead silicate glass with 50 mol% PbO.

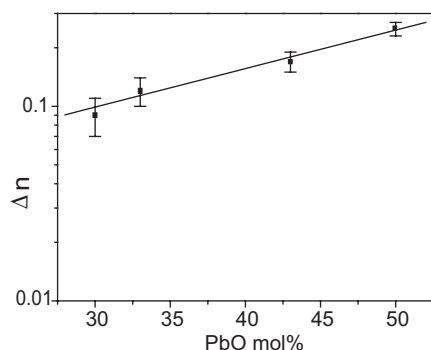


Fig. 3. The relationship between refractive index changes and PbO contents.

density. We could not measure the index of lead silicate glasses by Brewster's angle because of these spots.

Fig. 4 is the absorption spectra of lead silicate glass with 43 mol% PbO after exposure to 266 nm laser with

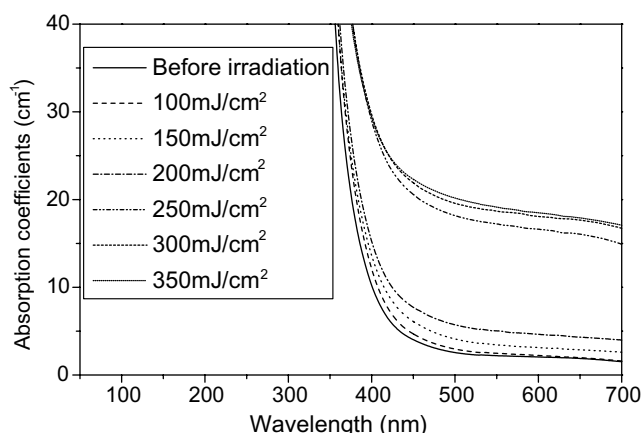


Fig. 4. The absorption spectra of lead silicate glass with 43 mol% PbO after irradiated by 266 nm laser with different energy density (the irradiation time is 1 min).

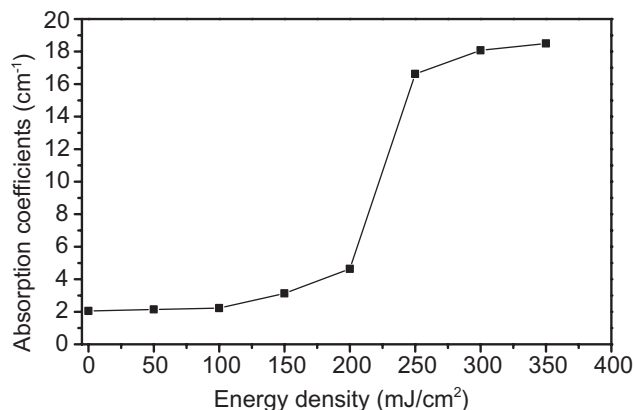


Fig. 5. The relationship between absorption coefficients at 600 nm and 266 nm laser energy density in lead silicate glass with 43 mol% PbO (the irradiation time is 1 min).

different energy density. It can be seen from Fig. 4 that the relationship of absorption coefficient and energy density of laser beam is non-linear. In order to see the relationship of absorption coefficient and energy density of laser beam clearly, we make a curve of absorption coefficient at 600 nm versus energy density (see Fig. 5). When the energy density was larger than 200 mJ/cm², the absorption coefficient increase evidently. When the energy density was smaller than 200 mJ/cm², the absorption coefficient increase slightly.

4. Discussion

There are two possibilities to induce the non-linear relationship between absorption coefficients and laser beam energy density. One possibility is that this is a two-photo absorption process. If this assumption was correct, the relationship of absorption coefficient and energy density should be second order polynomial. Another possibility is that this is a thermal process. Exposure to 266 nm laser beam, the temperature of local region of the lead silicate glasses may increase. The increasing of local temperature can cause the structural change and make the absorption coefficient increasing evidently. If this assumption was correct, the energy density should have a threshold value. Exposure to 266 nm laser beam with the energy density lower than the threshold value, the local temperature of lead silicate glass is not high enough to cause the structural change. Only when the energy density of the 266 nm laser beam is larger than the threshold value, the local temperature of lead silicate glass is high enough to cause the structural change and make the absorption coefficient increasing. We think the later possibility is corresponding to our experimental results.

Contardi et al. directly wrote channel waveguides in lead silicate glass by irradiating with a CW frequency doubled argon ion laser at 244 nm [11]. They also found that only the high power laser beam could write channel waveguide, which meant that the index change Δn was positive. The waveguide loss measured with He–Ne laser was great larger than the bulk loss of the glass, which meant that the absorption coefficient increased evidently. But the low power laser beam could not write channel waveguide, which meant that the index change Δn was not positive. They also thought that irradiating the lead silicate glass with 244 nm laser at high power was a heating process.

The absorption spectra of lead silicate glasses had different changes after exposure to the UV laser beam with different energy density. This showed that the micro-structural of lead silicate glasses had different changes after exposure to the UV laser beam with different energy density. Exposure to the UV laser beam with low energy density, the index change was negative. Exposure

to the UV laser beam with high energy density, the index change was positive and the absorption coefficient increased evidently. The increasing of the absorption coefficient might be induced by the increasing of local temperature of lead silicate glasses.

5. Conclusion

A frequency-quadrupled output of a Q-switched YAG laser (266nm, 10Hz repetition rate) was used to irradiate lead silicate glasses with different lead concentrations (from 30mol% PbO to 50mol% PbO). The refractive index of lead silicate glasses decreased after irradiation at low energy density (50mJ/cm²). The largest refractive index change Δn was -0.25 ± 0.04 in lead silicate glasses with 50mol% PbO. UV-visible absorption spectra of the lead silicate glass with 43mol% PbO were measured after exposure to the 266nm laser beams with energy density from 50mJ/cm² to 350mJ/cm². The absorption coefficients in visible wavelength increased suddenly when the energy density of the laser beam was larger than a threshold value. This showed that the structural changes of lead silicate glasses were different after exposure to the 266nm laser beam with different energy densities.

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